Research Methodology

# Week 4 Assignment: Questions, Answers and Explanations

### Week 4: Technical Presentation and Writing

NPTEL course noc26_ge79 (SWAYAM), 10 questions, 1 point each

Correct answers are the NPTEL portal's accepted answers. For every option the table gives a verdict and the reason, and each question ends with a short explanation of the correct answer. This week covers titles, images, tables, sentence style and document structure.

## Answer key at a glance

| Question | Type | Correct option(s) | Your portal score |
|---|---|---|---|
| **1** | Select all that apply | **C, D** | 1 (full marks) |
| **2** | Single answer | **B** | 1 (full marks) |
| **3** | Single answer | **C** | 1 (full marks) |
| **4** | Select all that apply | **C, D** | 1 (full marks) |
| **5** | Single answer | **A** | 1 (full marks) |
| **6** | Single answer | **A** | 1 (full marks) |
| **7** | Select all that apply | **A, B** | 1 (full marks) |
| **8** | Select all that apply | **A, B, C** | 1 (full marks) |
| **9** | Single answer | **B** | 1 (full marks) |
| **10** | Single answer | **B** | 1 (full marks) |

## Questions with explanations

### Question 1 of 10

**The title of a paper is 'Battery design and its impact'. This title**

*Select all that apply.*

| Option | Statement | Verdict | Reason |
|---|---|---|---|
| **A** | Identifies the area of research clearly | **Wrong** | "Battery" is too broad: it does not say which battery, application or kind of design. |
| **B** | Indicates what is unique about the research | **Wrong** | Nothing in the title says what is new. |
| **C** | **Does not identify the area of the research clearly** | **Correct** | Design of what, and impact on what, is left open. |
| **D** | **Does not indicate what is unique about the research** | **Correct** | The title could describe thousands of papers. |

**Correct answer: C and D.**

**Explanation:** A good title names the area precisely and says what is new. "Battery design and its impact" is vague on both counts, so it fails to identify the area and fails to show what is unique.

### Question 2 of 10

**In images**

*Single answer.*

| Option | Statement | Verdict | Reason |
|---|---|---|---|
| **A** | High magnification implies high resolution | **Wrong** | Enlarging a blurry image gives high magnification without extra detail. |
| **B** | **High magnification does not necessarily imply high resolution** | **Correct** | Magnification is how large the image is drawn; resolution is how much fine detail can be told apart. |
| **C** | Low magnification implies high resolution | **Wrong** | There is no such link between the two. |
| **D** | High resolution implies low magnification | **Wrong** | A high-resolution image can be shown at any magnification. |

**Correct answer: B.**

**Explanation:** Magnification and resolution are independent. You can enlarge an image a lot (high magnification) and still see no finer detail (low resolution).

## Question 3 of 10

**Tables are preferred over graphs**

*Single answer.*

| Option | Statement | Verdict | Reason |
|---|---|---|---|
| **A** | Always | **Wrong** | Graphs are better for showing trends. |
| **B** | Never | **Wrong** | Tables are better when exact values matter. |
| **C** | **Only when exact values need to be indicated** | **Correct** | A table lets the reader read precise numbers. |
| **D** | Only when trends in data need to be indicated | **Wrong** | Trends are shown better by a graph. |

**Correct answer: C.**

**Explanation:** Use a table when the reader needs exact values and a graph when the reader needs to see a trend or compare shapes at a glance.

## Question 4 of 10

**The length of a document**

*Select all that apply.*

| Option | Statement | Verdict | Reason |
|---|---|---|---|
| **A** | Indicates how much you know | **Wrong** | A long document may simply be padded. |
| **B** | Indicates how little you know | **Wrong** | A short document may be precise. |
| **C** | **Does not indicate anything about your knowledge** | **Correct** | Length depends on style and purpose, not on the author's knowledge. |
| **D** | **Does not indicate anything about the reader's knowledge** | **Correct** | What the reader already knows is independent of the document's length. |

**Correct answer: C and D.**

**Explanation:** Length is not a measure of anybody's knowledge. It says nothing about what the author knows and nothing about what the reader knows.

## Question 5 of 10

**The title of a paper should indicate the area of work as well as what is unique about the work.**

*Single answer.*

| Option | Statement | Verdict | Reason |
|---|---|---|---|
| **A** | **True** | **Correct** | A title should tell the reader where the work belongs and what is new. |
| **B** | False | **Wrong** | A title that omits these gives the reader little reason to read on. |

**Correct answer: A.**

**Explanation:** Readers use the title to decide whether to read the paper, so it should state the area and the unique contribution, as Question 1 showed.

## Question 6 of 10

**In scientific documents sentences should be relatively short and to the point.**

*Single answer.*

| Option | Statement | Verdict | Reason |
|---|---|---|---|
| **A** | **True** | **Correct** | Short direct sentences carry one idea and are hard to misread. |
| **B** | False | **Wrong** | Long sentences make technical text harder to follow. |

**Correct answer: A.**

**Explanation:** Technical readers want precision. Short sentences that make one point each reduce ambiguity.

## Question 7 of 10

**In an article, the image of a car engine must**

*Select all that apply.*

| Option | Statement | Verdict | Reason |
|---|---|---|---|
| **A** | **Show the level of detail required for the discussion in the article** | **Correct** | An image should show as much detail as the text discusses, no more and no less. |
| **B** | **Have a marker bar indicating scale** | **Correct** | The printed size can change, so a scale bar tells the real size. |
| **C** | Be in colour and not in black and white | **Wrong** | Colour is needed only if it carries information; it is not a rule. |
| **D** | Be in black and white and not in colour | **Wrong** | Black and white is not a rule either. |

**Correct answer: A and B.**

**Explanation:** A figure must serve the discussion, so it shows the needed level of detail and carries a scale bar. Whether it is in colour depends on whether colour helps the reader.

## Question 8 of 10

**Strategies one can use in the middle of a document include**

*Select all that apply.*

| Option | Statement | Verdict | Reason |
|---|---|---|---|
| **A** | **Chronological strategy** | **Correct** | Arrange the content in time order. |
| **B** | **Spatial strategy** | **Correct** | Arrange it by position, for example part by part of a machine. |
| **C** | **Following a variable** | **Correct** | Follow one quantity through the work. |
| **D** | Suspense strategy | **Wrong** | Suspense is a device of fiction; technical readers want the point early. |

**Correct answer: A and B and C.**

**Explanation:** The body of a technical document can be organised by time, by space or by following a variable. Building suspense is not used in technical writing.

## Question 9 of 10

**While working with electricity only the voltage poses a danger to the users**

*Single answer.*

| Option | Statement | Verdict | Reason |
|---|---|---|---|
| **A** | True | **Wrong** | Voltage is not the only danger. |
| **B** | **False** | **Correct** | The harm to the body comes from the current flowing through it, which depends on voltage and resistance. |

**Correct answer: B.**

**Explanation:** It is the current through the body that does the damage. A modest voltage with a low-resistance path can still push a dangerous current, so voltage alone is not the whole story.

## Question 10 of 10

**The sentence**

“Pouring rain results in clogged drains, and slippery roads. These clogged drains spill further water on to the roads.”

**Use of the word ‘these’ in the previous sentence is ambiguous.**

*Single answer.*

| Option | Statement | Verdict | Reason |
|---|---|---|---|
| **A** | True | **Wrong** | "These" is followed by the noun it refers to, so nothing is unclear. |
| **B** | **False** | **Correct** | "These clogged drains" clearly points to the clogged drains in the first sentence. |

**Correct answer: B.**

**Explanation:** A pronoun such as "this" or "these" is ambiguous when it stands alone and could refer to more than one earlier item. Here it is followed by the noun "clogged drains", so the reference is clear.